



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First / Second Semester B.Tech (CBCS) Degree Examinations

Summer Semester – July 2024

CS1005 -1 INTRODUCTION TO PYTHON PROGRAMMING

(AD,AM,BT,CB,CC,CS,CV,ME,IS,RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What is the primary purpose of a variable in Python?**
A) To store data
B) To print output
C) To perform calculations
D) To define functions
2. **What does the range(3) function in a for loop represent in Python?**
A) It creates a list with values [0, 1, 2, 3].
B) It creates a range object from 0 to 3.
C) It creates a list with values [0, 1, 2].
D) It's an invalid Python expression
3. **In Python, what is the output of the following code?**

```
x = 10
for i in range (5):
    x += i
print(x)
```


A) 20
B) 15
C) 14
D) 21
4. **In Python, what is the purpose of the continue statement in a loop?**
A) It terminates the loop.
B) It skips the current iteration and continues with the next one.
C) It restarts the loop from the beginning.
D) It prints the current loop iteration.
5. **To find the index of the first occurrence of a value in a list, which list method should be used?**
A) index()
B) find()
C) search()
D) locate()
6. **Which Python operator is used to access a value associated with a specific key in a dictionary?**
A) :
B) .
C) ->
D) []
7. **What is the purpose of the dict.keys() method in Python?**
A) It returns the values of the dictionary.
B) It returns a list of all the keys in the dictionary.
C) It returns the keys associated with specific values.
D) It returns the length of the dictionary.
8. **What is the primary purpose of the with statement when opening files in Python?**
A) To create a new file.
B) To open a file in read-only mode.
C) To ensure proper file closure and resource management.
D) To check if the file exists.
9. **In Python, how do you define a function?**
A) Using the create keyword.
B) Using the function keyword.
C) Using the define keyword.
D) Using the def keyword.

10. What is the advantage of using a lambda function over a regular function in Python?
- A) Lambda functions can handle exceptions more effectively. B) Lambda functions can be called from other programming languages.
- C) Lambda functions are more memory-efficient and can be used for short tasks without defining a separate function. D) Lambda functions have built-in error handling.
11. What is the purpose of the *args and **kwargs parameters in Python function definitions?
- A) They are used to specify the function's name and arguments. B) They allow the function to accept a variable number of positional and keyword arguments.
- C) They are used to raise exceptions in the function. D) They define the function's return type.
12. What does the count() method in Python return when counting occurrences of a substring in a string?
- A) The index of the first occurrence. B) The number of occurrences.
- C) The position of the last occurrence. D) The substring itself.
13. Which of the following is the truncation division operator in Python?
- A) | B) //
- C) / D) %
14. How do you convert a string to lowercase in Python?
- A) my_string.lowercase() B) my_string.to_lowercase()
- C) my_string.lower() D) my_string.convert_to_lowercase()
15. How can you efficiently join a list of strings into a single string in Python, assuming there is a large number of strings to be joined?
- A) Using a for loop. B) Using the str.join() method.
- C) Using a list comprehension. D) None of the above
16. How can you convert a string to a list of characters in Python?
- A) Using the str.split() method. B) By casting the string to a list.
- C) By using the str.to_list() function D) By using a for loop and appending characters one by one.
17. What is the purpose of the os module in Python?
- A) To interact with the operating system and file system B) To create GUI applications
- C) To perform mathematical operations D) To work with databases
18. What will be the output of the following code snippet?
- ```
for i in [1, 2, 3, 4] [::-1]:
 print(i)
```
- A) 4 3 2 1 B) Error
- C) 1 2 3 4 D) None of the mentioned
19. In Python, how can you create a new directory?
- A) Use the mkdir() method B) Use the new\_dir() method
- C) Use the create\_directory() function D) Use the os.makedirs() function
20. What is the purpose of the shutil module in Python?
- A) To create text files B) To move or copy files and directories.
- C) To read and write binary data D) To compress and decompress files

### PART - B: DESCRIPTIVE ANSWER QUESTIONS

|       | Unit – I                                                                                                                                         | Marks | BT* | CO* | PO* |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----|
| 1. a) | What is an Algorithm? Write an algorithm and flowchart for a program that converts temperature from Celsius to Fahrenheit $F = (C * 9/5) + 32$ . | 5     | L*2 | 1   | 1,2 |
| b)    | Write and Explain step by step procedure to display numbers from 1 to 5 using While loop.                                                        | 6     | L2  | 1   | 2   |

|                   |                                                                                                                                                                                                                                                  |   |    |   |     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|---|-----|
| c)                | Write a python program to find the sum of digits and number of occurrence of digit in a number.                                                                                                                                                  | 5 | L3 | 1 | 2   |
| 2. a)             | Explain the various steps involved in program development with a neat diagram                                                                                                                                                                    | 6 | L2 | 1 | 1   |
| b)                | What is a variable? What are the rules for writing an identifier?                                                                                                                                                                                | 5 | L1 | 1 | 1   |
| c)                | Write a python program to check whether the given string is palindrome or not.                                                                                                                                                                   | 5 | L3 | 1 | 2   |
| 3. a)             | Explain different types of conditional Control statements with its syntax and example program                                                                                                                                                    | 6 | L2 | 1 | 2   |
| b)                | How does RAM differ from secondary storage in terms of data storage and access?                                                                                                                                                                  | 4 | L2 | 1 | 1   |
| c)                | List and Explain Comparison operator with example program.                                                                                                                                                                                       | 6 | L2 | 1 | 2   |
| <b>Unit – II</b>  |                                                                                                                                                                                                                                                  |   |    |   |     |
| 4. a)             | Write a Python function that finds the maximum and minimum values in a list without using the built-in max () and min () functions.                                                                                                              | 6 | L3 | 3 | 2,3 |
| b)                | Explore the idea of recursive functions. Write a program to print Fibonacci sequences using recursion.                                                                                                                                           | 5 | L2 | 3 | 2   |
| c)                | Write a Python program to convert a list of tuples into a dictionary using a function..                                                                                                                                                          | 5 | L3 | 2 | 2   |
| 5. a)             | What is a List? Write Python code to replace the third element in a list with a new value using indexing concept.                                                                                                                                | 5 | L2 | 2 | 1,2 |
| b)                | Write a Python function that receives two dictionaries as arguments and merges them into a new dictionary. If there are common keys, combine their values.                                                                                       | 6 | L3 | 3 | 2   |
| c)                | Describe the benefits of using named arguments when calling functions. Provide an example to illustrate the use of named arguments.                                                                                                              | 5 | L2 | 3 | 2   |
| 6. a)             | Write a Python program to read a list of n integers (positive as well as negative). Create 2 new lists, one having all positive numbers and others having all negative numbers from the given list. Print all 3 lists.                           | 5 | L3 | 2 | 2,3 |
| b)                | Explain the distinction between instance variables and class variables in Python.                                                                                                                                                                | 5 | L2 | 3 | 1,2 |
| c)                | What are formal arguments and actual arguments in function definitions? How do they relate to function parameters and arguments passed during function calls?                                                                                    | 6 | L2 | 3 | 2   |
| <b>Unit – III</b> |                                                                                                                                                                                                                                                  |   |    |   |     |
| 7. a)             | Discuss the differences between slicing and substring extraction from a string. Provide examples to illustrate.                                                                                                                                  | 8 | L3 | 4 | 2   |
| b)                | Create a Python program that creates a CSV file named "data.csv" and writes the following data into it:<br><i>Name, Age, City</i><br><i>John, 25, New York</i><br><i>Emily, 30, Los Angeles</i><br>Then, read the CSV file and display the data. | 8 | L3 | 5 | 3   |
| 8. a)             | Explain any four string manipulation function with an example program for each?                                                                                                                                                                  | 8 | L2 | 4 | 2   |
| b)                | How do you open a text file in Python for reading, and what is the most common mode used for this operation?                                                                                                                                     | 8 | L3 | 5 | 1,2 |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

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